

SHCHUCHINSK, Kazakhstan

WMO: 289840

Lat: 52.9251N

Lon: 70.2038E

Elev: 1296

StdP: 14.02

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.8	-21.7	-33.5	0.9	-27.3	-27.3	1.3	-21.5	22.8	20.3	20.4	19.0	2.2	300	0.615

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.8	85.8	62.9	82.5	62.2	79.3	61.3	66.4	79.0	65.0	76.8	63.6	74.8	7.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.4	88.3	70.0	60.9	83.8	69.1	59.3	79.1	68.0	31.8	79.5	30.7	77.3	29.6	74.5	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	22.7	18.9	16.9	DB	-33.4	90.9	5.1	3.7	-37.1	93.6	-40.1	95.8	-42.9	97.9	-46.6	100.6
(5)				WB	-33.2	69.3	5.3	1.9	-37.0	70.7	-40.1	71.8	-43.0	72.8	-46.8	74.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	35.7	2.0	4.5	19.7	40.2	53.4	62.7	64.0	62.0	51.2	38.4	21.2	7.4
	DBStd	24.91	12.89	12.19	10.68	10.68	9.21	7.70	6.42	7.34	8.90	9.44	13.16	12.85
	HDD50	6752	1488	1274	939	323	72	6	0	3	89	373	865	1320
	HDD65	10887	1953	1694	1404	745	371	131	96	147	422	824	1315	1785
	CDD50	1545	0	0	0	28	179	386	435	376	126	15	0	0
	CDD65	200	0	0	0	0	12	61	65	54	8	0	0	0
	CDH74	2262	0	0	0	11	193	711	624	587	135	1	0	0
	CDH80	577	0	0	0	0	33	197	152	162	33	0	0	0
Wind	WSAvg	6.2	5.2	6.1	6.8	6.9	7.1	6.4	5.5	5.4	5.7	6.2	6.8	6.3
Precipitation	PrecAvg	13.8	0.6	0.5	0.7	0.8	1.3	1.6	3.3	1.5	0.9	1.0	0.9	0.7
	PrecMax	20.3	1.6	1.1	1.3	2.1	3.0	3.3	7.3	3.8	2.0	2.1	2.1	2.0
	PrecMin	7.5	0.2	0.2	0.1	0.1	0.1	0.0	0.8	0.2	0.0	0.1	0.3	0.2
	PrecStd	3.6	0.3	0.3	0.4	0.5	0.7	0.9	1.9	1.0	0.5	0.5	0.5	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.5	32.4	46.4	74.7	84.5	89.6	89.7	88.8	84.5	69.9	46.7	33.0
		MCWB	30.2	30.2	38.4	53.7	59.7	63.1	64.9	63.8	58.6	53.2	41.5	31.3
	2%	DB	27.8	28.8	39.6	67.6	79.2	85.1	84.1	84.5	77.6	63.3	42.6	30.8
		MCWB	26.4	27.3	35.6	51.3	57.3	62.3	63.9	63.2	56.7	48.4	39.2	29.5
	5%	DB	24.3	25.8	36.4	62.6	75.3	81.8	80.2	80.5	72.1	58.4	38.6	28.0
		MCWB	23.1	24.6	33.5	49.0	55.7	61.7	63.6	62.7	54.7	46.1	35.9	26.6
	10%	DB	21.0	22.6	33.8	57.5	71.2	78.1	77.1	76.5	66.6	52.8	35.1	24.6
		MCWB	19.7	21.5	31.6	46.6	54.5	60.7	62.7	61.3	52.8	44.0	32.9	23.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.5	30.8	39.9	56.1	62.8	67.2	68.7	67.9	61.8	53.1	43.3	32.1
		MCDB	31.4	31.8	45.0	70.0	76.2	81.9	79.9	80.8	78.9	65.4	45.7	32.7
	2%	WB	26.7	27.4	36.1	52.8	59.7	65.2	66.7	65.7	58.5	50.2	39.4	29.6
		MCDB	27.7	28.7	38.9	64.6	73.9	79.1	78.6	78.0	72.6	60.9	42.4	30.6
	5%	WB	23.2	24.7	33.9	49.9	57.6	63.6	65.4	64.1	56.2	46.9	36.1	26.7
		MCDB	24.2	25.8	36.0	60.5	71.4	76.8	76.6	76.1	69.5	56.2	38.4	27.9
	10%	WB	19.9	21.5	31.7	47.0	55.7	62.1	64.0	62.4	53.7	44.5	33.2	23.3
		MCDB	21.0	22.6	33.8	57.0	68.5	74.9	73.8	74.0	64.2	52.1	35.0	24.4
Mean Daily Temperature Range	5% DB	MDBR	15.4	16.6	16.1	19.3	22.5	21.0	18.8	20.4	20.2	16.1	11.9	14.1
		MCDBR	13.0	14.2	14.3	26.9	28.4	26.1	24.7	26.5	28.5	26.0	13.9	13.6
		MCWBR	12.7	13.4	11.2	14.4	12.9	10.7	10.0	10.9	12.4	14.5	10.8	13.3
	5% WB	MCDBR	12.8	13.2	12.7	25.2	24.8	22.4	20.6	22.8	25.1	22.6	12.5	13.7
		MCWBR	12.6	12.6	10.6	14.8	13.0	10.3	9.7	11.0	12.3	13.6	10.4	13.5
Clear-Sky Solar Irradiance	taub	0.278	0.306	0.361	0.398	0.394	0.399	0.421	0.409	0.357	0.338	0.311	0.278	
	taud	2.083	2.055	1.994	2.155	2.242	2.306	2.235	2.250	2.355	2.303	2.155	2.070	
	Ebn at Noon	214	246	254	263	271	271	261	257	259	236	199	187	
	Edh at Noon	23	33	43	42	41	39	41	38	30	26	22	20	
All-Sky Solar Radiation	RadAvg	386	745	1207	1472	1817	1929	1769	1532	1121	596	291	256	
	RadStd	28	28	123	144	158	151	135	117	78	80	39	35	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page